Lecture 6.3: Machine Learning in Economic Research

AREC 570 / ECON 530 · Unit 6

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Before class

Assigned

- Athey and Imbens, “The State of Applied Econometrics: Causality and Policy Evaluation” (Athey and Imbens 2017)
- Athey and Imbens, “Machine Learning Methods That Economists Should Know About” (Athey and Imbens 2019)
- Mullainathan and Spiess, “Machine Learning: An Applied Econometric Approach” (Mullainathan and Spiess 2017)

Readings and course handouts are available on Canvas.

Example papers in these slides

- Overview for agribusiness: Maruejols et al. (2025)
- Prediction: Zhao et al. (2025)
- Prediction within a larger analysis: Li and Yu (2025); Belloni et al. (2014)
- Heterogeneous treatment effects: Stetter et al. (2022)
- Further reading: Bittmann’s guide from OLS to causal machine learning (Bittmann 2025)

Lectures 6.1 and 6.2 estimated parameters; machine learning targets prediction

- The designs in 6.1 and the models in 6.2 recover parameters: a treatment effect, a price coefficient, an elasticity
- Supervised machine learning addresses a different objective: the most accurate prediction of y given x for observations outside the estimation sample
- In Mullainathan and Spiess's words, machine learning "not only provides new tools, it solves a different problem"
- This lecture covers where these methods came from, the three roles they play in empirical economics, and how a methods section reports them

Source: Mullainathan and Spiess (2017), pages 87 to 88.

Background

Machine learning entered applied economics within the last decade

Period	Development
1984 to 2001	Regression trees (Breiman et al., 1984), the LASSO (Tibshirani, 1996), and random forests (Breiman, 2001)
1990s	Neural networks studied in econometrics, but they “did not catch on at the time”
2001	Breiman’s “two cultures”: statistics relied on data models, while an algorithmic culture grew in other fields
2014	Early reviews for economists: Varian; Einav and Levin; Belloni, Chernozhukov, and Hansen
2017 to 2019	Mullainathan and Spiess; causal forests (Wager and Athey); Athey and Imbens’s review for economists
2020 to 2025	Reviews and special issues in agricultural and applied economics: Storm et al. (2020); Maruejols et al. (2025)

Sources: Athey and Imbens (2019), pages 686 to 688, 696 to 698, and 701; Mullainathan and Spiess (2017), page 87; Storm et al. (2020); Maruejols et al. (2025).

Why economists adopted machine learning, and why adoption was slow

- **New data:** many units, many variables per unit, and unstructured sources such as satellite images and text
- **Prediction questions:** some policy problems require an accurate forecast and no causal parameter
- **Many candidate variables:** the analyst must choose controls or functional forms from a large set
- **Why adoption was slow:** economics journals emphasize formal properties of estimators, such as consistency and asymptotic normality, which early machine learning methods did not provide
- **What changed:** recent work established inference for adapted methods, including double selection and causal forests

Sources: Athey and Imbens (2019), pages 686 to 687; Mullainathan and Spiess (2017), pages 88, 98, and 102.

Prediction and parameter estimation are evaluated by different criteria

	Prediction, $\widehat{y}$	**Parameter estimation, $\widehat{\beta}$**
Target	The outcome for a new observation	The effect of a change in x on y
Evaluation	Accuracy on data not used in estimation	Identification: the source of variation and its assumptions
Verifiability	Directly testable: predictions and outcomes are both observed	Rests on assumptions about the data-generating process
Typical methods	LASSO, random forests, neural networks	OLS, IV, difference in differences, structural models

Source: Mullainathan and Spiess (2017), pages 88 and 92.

Machine learning plays three roles in empirical economics

Role	What machine learning does	Example
1. Prediction as the objective	Predicts an outcome, such as household food insecurity	Zhao et al. (2025)
2. Prediction within a larger analysis	Selects the variables that enter an economic model or a treatment-effect regression	Li and Yu (2025); Belloni et al. (2014)
3. Heterogeneous treatment effects	Estimates how a treatment effect varies across farms or households	Stetter et al. (2022)

In roles 2 and 3, identification still comes from the research design. Machine learning does not supply exogenous variation.

Sources: Mullainathan and Spiess (2017); Athey and Imbens (2019), pages 686 to 688; Maruejols et al. (2025).

Core concepts

Out-of-sample performance: training data, test data, and cross-validation

- A flexible model can fit noise in its estimation sample, so in-sample fit overstates predictive performance (overfitting)
- **Training and test data:** the model is estimated on one subset and evaluated on a held-out subset
- **Cross-validation:** the training data are divided into K folds; the model is estimated on $K - 1$ folds and evaluated on the remaining fold, in rotation, to select tuning parameters

Paper	Sample split
Zhao et al. (2025)	70% training (756 households), 30% test (324 households)
Li and Yu (2025)	10-fold cross-validation to select the LASSO penalty

Sources: Mullainathan and Spiess (2017), pages 89 to 92; Bittmann (2025), Section 3.

Regularization: the LASSO and random forests

LASSO adds a penalty on the absolute size of the coefficients to the least squares objective:

$$\min_{\beta} \sum_{i=1}^{n} \left(y_i - x_i' \beta \right)^2 + \lambda \sum_{j=1}^{p} |\beta_j|$$

- With $\lambda = 0$, the estimator is OLS; as λ increases, coefficients shrink and some are set exactly to zero
- The LASSO therefore performs variable selection; λ is chosen by cross-validation

Random forests average the predictions of many regression trees, each estimated on a random subsample, which captures nonlinearities and interactions without specifying them in advance.

Sources: Mullainathan and Spiess (2017), pages 90 to 97; Athey and Imbens (2019), pages 690 and 695 to 699; Bittmann (2025), Section 4.

Role 1: Prediction as the objective

Zhao et al. (2025) predict food insecurity among rural households in China

- **Objective:** classify rural households as calorie poor
- **Data:** 1,080 households in Jilin, Shandong, and Hubei; 3-day food consumption records; 61.1% of households are calorie poor
- **Methods:** random forest and LASSO, compared with a logit benchmark; 70% training and 30% test samples

Model (test sample)	Accuracy
Random forest	65.7%
LASSO	61.1%
Logit benchmark	58.9%
Classify every household as calorie poor	61.1%

Accuracy is informative only relative to a benchmark; the final row is the majority-class rate.

Source: Zhao et al. (2025), Sections 3 to 5 and Table 3. The final row is our calculation from the sample share.

Predictive importance does not identify causal effects

- Zhao et al. rank predictors; household income and access to food markets rank highest in both models
- A variable may predict well because it is correlated with an unobserved determinant of the outcome
- LASSO selections are unstable: across ten random partitions of the same data, Mullainathan and Spiess find that the selected variables change from one partition to the next
- Stetter et al. caution that their variable importance measure “should not be interpreted in a causal fashion”

A causal claim about income would require an identification strategy of the kind discussed in Lecture 6.1.

Sources: Zhao et al. (2025), Section 5.2; Mullainathan and Spiess (2017), page 97, Figure 2; Stetter et al. (2022), page 740, footnote 17.

Role 2: Prediction within a larger analysis

Li and Yu (2025) use the LASSO to select attributes before estimating willingness to pay

- **Setting:** a choice experiment with 100 consumers in Göttingen; apples vary in origin, size, production method, and price
- **First stage, LASSO logit:** 10-fold cross-validation selects λ, with price exempt from the penalty; the LASSO excludes **Organic** and **Japan**, interpreted as attribute non-attendance
- **Second stage, conditional logit:** the random utility model from Lecture 6.2, with willingness to pay $= -\beta_k / \beta_{\text{price}}$

Willingness to pay (€ per kg), relative to China	All attributes	After LASSO
France	1.20	1.53
Italy	1.42	1.40
Small apples	1.30	1.41
Organic	−0.04	excluded

Source: Li and Yu (2025), Sections 2 and 3, Tables 4, 6, and 7.

Belloni, Chernozhukov, and Hansen (2014) use the LASSO to select control variables

With a treatment d_i and many candidate controls x_i, **double selection** proceeds in three steps:

1. LASSO of the treatment d_i on x_i, retaining controls that predict treatment
2. LASSO of the outcome y_i on x_i, retaining controls that predict the outcome
3. OLS of y_i on d_i and the **union** of the two sets of controls

- Selecting controls only for their ability to predict y can omit a confounder that strongly predicts treatment
- **Application:** abortion and crime; 48 states over 12 years (576 observations), 251 candidate controls, of which the LASSO retains 8 to 14
- **Identification:** the estimate is causal only if treatment is as good as randomly assigned conditional on the controls

Source: Belloni et al. (2014), Sections 1, 2.2, and 7.

Role 3: Heterogeneous treatment effects

Stetter et al. (2022) estimate farm-level effects of agri-environment schemes with a causal forest

The estimand is the **conditional average treatment effect**:

$$\tau(x) = E\left[\, Y_i(1) - Y_i(0) \mid X_i = x \,\right]$$

- **Data:** 2,758 Bavarian farms in 2014, of which 1,641 participated in an agri-environment scheme; more than 130 farm and county characteristics
- **Method:** forests estimate participation probabilities and expected outcomes; a causal forest then estimates $\tau(x)$ for each farm, with tuning parameters chosen by cross-validation
- **Results:** fertiliser expenditure falls significantly for 33% of farms (by €14 per hectare among them); land-use diversity rises for 55%; effects on pesticide use and greenhouse gas emissions are rarely significant

Source: Stetter et al. (2022), Sections 3 to 5 and Table 3.

Causal forests rely on the same identifying assumptions as the designs in Lecture 6.1

- **Selection on observables:** conditional on farm characteristics, participation is as good as randomly assigned
- **Overlap:** every farm has a positive probability of each treatment status; estimated propensities range from 0.27 to 0.86, and 23 farms without overlap are excluded
- **The authors' argument:** a large set of characteristics allows the forest to proxy for unobserved factors, such as a farmer's environmental attitude
- **The acknowledged limit:** if a distinct group of confounders is absent from the data, "our estimates might still be biased"
- **Sensitivity analysis:** placebo tests and simulated confounders

Source: Stetter et al. (2022), Sections 4.3 and 5.1.

Reporting a machine learning method

What a methods section reports when it uses machine learning

Item	Example
1. The role: prediction, a step in a larger analysis, or heterogeneous effects	Zhao et al.: prediction is the objective
2. The target, the inputs, and a benchmark model	Calorie poverty; household and market characteristics; logit
3. The sample split	70% training, 30% test
4. The choice of tuning parameters	Cross-validation; number of trees
5. Out-of-sample performance	Accuracy, sensitivity, specificity, area under the ROC curve
6. For roles 2 and 3: the identifying assumption and its checks	Stetter et al.: selection on observables, overlap, placebo tests
7. Software and versions	Li and Yu: the `glmnet` package in R

Sources: Zhao et al. (2025); Li and Yu (2025); Stetter et al. (2022); Athey and Imbens (2019), pages 686 to 687.

What to consider for your proposal

- **The role:** does the question concern predicting an outcome, selecting variables, or how an effect varies?
- **The claim:** does the conclusion require a causal effect? If so, what is the design, and where does machine learning enter it?
- **The benchmark:** which simpler model or rule should the method outperform?
- **The evidence:** which out-of-sample measure, or which identification check, will the paper report?

References

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